

Research Summary: Black Holes, Galaxies, and Transients from Space-based Spectroscopy and Imaging

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My research uses space-based spectroscopy and imaging to investigate how black holes grow and how galaxies form their stars. A central challenge is to distinguish physical diversity from the selection effects that shape our observed samples. During my PhD, I developed quasar-selection methods for HST slitless spectra and a simulation-based pipeline for the Chinese Space Station Telescope (CSST). My subsequent research extends this foundation to JWST: measuring the abundance of faint star-forming galaxies, probing the central engines of little red dots (LRDs), and identifying distant transients. Across these projects, I connect source discovery, quantitative characterization of survey selection, and physical interpretation. The five first-author studies below define this progression.

1. PHD FOUNDATION: FINDING AND CHARACTERIZING QUASARS WITH SLITLESS SPECTRA

HST: separating broad-line AGN from spatially extended galaxies. An apparent broad line in a slitless spectrum can arise from either high-velocity gas or the spatial extent of a galaxy. I developed a selection procedure that addresses this ambiguity through two-dimensional forward modeling, rather than relying on the extracted line width alone (Pang et al. 2025a). After screening more than 6,000 emission-line sources in 3D-HST, I used a modified GRIZLI model to evaluate the evidence for a quasar component. Tests recovered about 90% of known, point-like quasars with detectable $H\alpha$ or $H\beta$, with an emission-line-galaxy contamination rate of about 5%.

The resulting sample contains **19 quasar candidates at $z = 0.12$ – 1.56** , including 12 with Chandra detections. Their broader rest-frame UV-color range and redder optical slopes relative to SDSS quasars demonstrate the value of selection without a quasar-color cut. These candidates improve the census at $z = 0.8$ – 1.6 in the 3D-HST fields. This work established a practical way to recover AGN missed by conventional color selection, while explicitly accounting for morphology and spectral contamination.

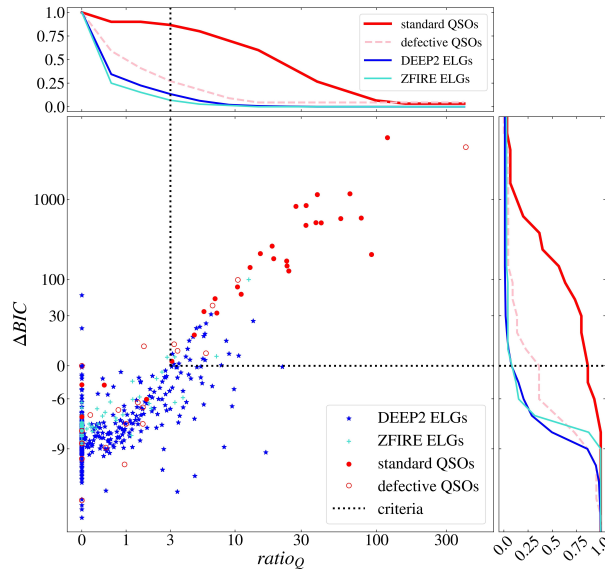


Figure 1. Quasar selection in 3D-HST. The selection diagnostic compares the strength of the fitted quasar component with the improvement in the Bayesian information criterion. This provides a way to assess evidence for a quasar component while accounting for spatially extended emission in slitless spectra. Figure from my PhD research; see Pang et al. (2025a).

CSST: taking quasar surveys to much larger samples. I developed a pipeline linking simulated grism images to spectral extraction, convolutional-neural-network classification, emission-line redshifts, and spectral fitting (Pang et al. 2025b). For mock sources with $18 < m_r < 22$, classification accuracy reaches 99%; about 90% of successfully classified quasars have precise redshifts, with $\sigma_{\text{NMAD}} = 0.002$. Recovered black-hole masses and Eddington ratios show scatters of 0.13 and 0.15 dex relative to the input values. These quantify recovery within the simulation, rather than the absolute accuracy of virial mass estimators. Under the adopted survey assumptions, the forecast is **approximately 0.9 million new quasars**. Together, these studies provide complementary methods for deep, small-area observations and large survey datasets.

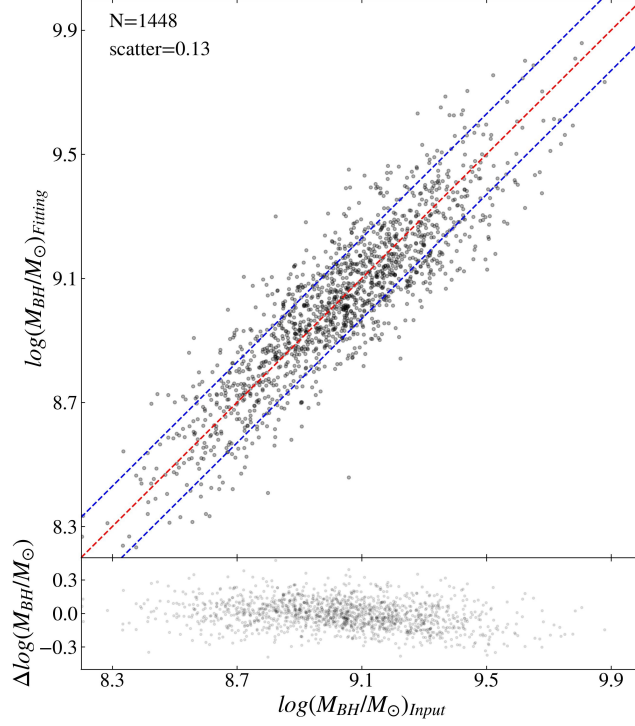


Figure 2. Black-hole mass recovery with CSST mock spectra. Recovered black-hole masses are compared with the input values used to generate the simulated quasars. The 0.13-dex scatter quantifies recovery within the simulation; it does not include the systematic uncertainty of single-epoch virial mass estimators. Figure from my PhD research; see Pang et al. (2025b).

2. JWST GALAXY POPULATIONS: THE FAINT END OF STAR FORMATION AT COSMIC NOON

The next step in my research was to turn deep slitless spectroscopy into a measurement of galaxy demographics. In my first-author GLASS-JWST study, I measured the $H\alpha$ luminosity function at $z \sim 1.3$ and $z \sim 2.0$, using NIRISS spectroscopy and gravitational magnification in the Abell 2744 field (Pang et al. 2026b). The scientific aim was to constrain the contribution of faint galaxies to star formation during cosmic noon, where the luminosity function is sensitive to incompleteness near the survey limit.

Starting from 99 spectroscopically selected $H\alpha$ emitters, the final luminosity-function analysis uses 53 and 29 distinct galaxies in the two redshift bins. I developed a source-by-source treatment of effective survey volume and completeness that incorporates the spatially varying depth and lensing magnification. Mock-line recovery tests quantify detection completeness and flux bias, while the analysis accounts for multiply imaged sources and cosmic variance. This work extends the emphasis of my PhD from recognizing individual sources to understanding the population that a survey can actually detect.

The observations reach intrinsic line luminosities of approximately $10^{40.5}$ and $10^{40.9}$ erg s^{-1} in the two bins, roughly an order of magnitude fainter than previous grism surveys. We measure faint-end slopes of

$$\alpha = -1.50^{+0.14}_{-0.08} \quad (z \sim 1.3), \quad \alpha = -1.60^{+0.17}_{-0.09} \quad (z \sim 2.0).$$

These measurements constrain the faint population directly, reducing reliance on an assumed slope when integrating the luminosity function. After applying the adopted dust and $[\text{N II}]$ corrections, we infer cosmic star-formation-rate densities of $0.097^{+0.015}_{-0.016}$ and $0.129^{+0.025}_{-0.030}$ $M_{\odot} \text{ yr}^{-1} \text{ Mpc}^{-3}$, respectively.

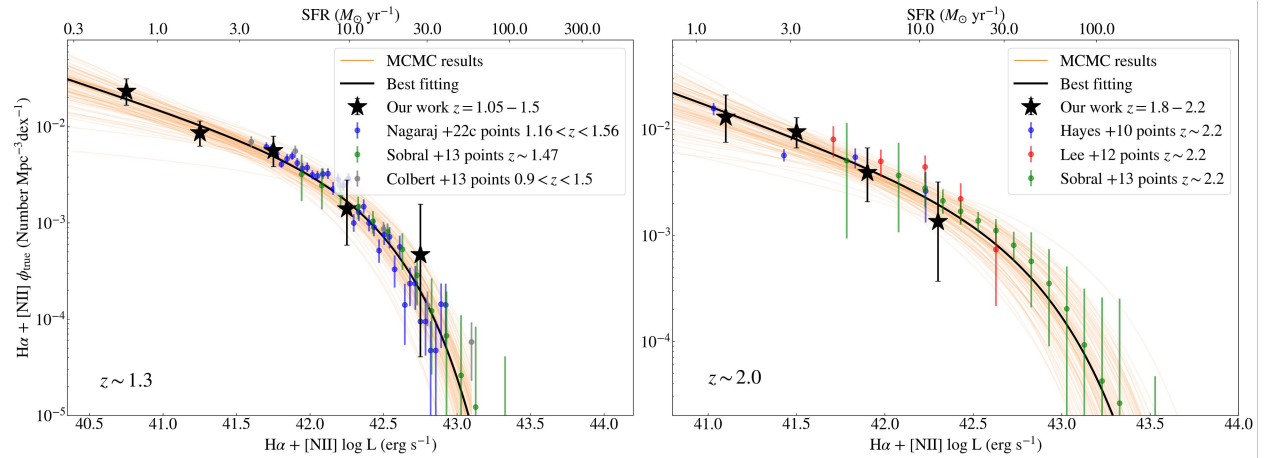


Figure 3. Reaching faint galaxies with JWST and gravitational lensing. Luminosity functions at $z \sim 1.3$ and $z \sim 2.0$, reproduced from Figure 5 of Pang et al. (2026b). Black stars show our measurements; black curves and orange realizations show the fitted Schechter functions and their uncertainty. As labeled on the axes, the plotted line luminosity is the unresolved $H\alpha + [\text{N II}]$ blend; the star-formation analysis subsequently applies $[\text{N II}]$ and dust corrections.

Scientific contribution and continuity. This project demonstrates how sensitivity, lensing, and a measured selection function can be combined to constrain galaxies below the reach of earlier grism surveys. It also provides a methodological connection between my quasar work and my broader interest in galaxy evolution: an observed source count becomes physically informative only when its accessible volume and detection probability are understood. The same approach can support future comparisons between emission-line-selected AGN, star-forming galaxies, and samples selected through broadband photometry.

3. JWST BLACK-HOLE PHYSICS: THE STRUCTURE AND EVOLUTION OF LITTLE RED DOTS

While my survey work asks which objects are missing from our samples, my LRD study asks what the newly revealed objects are. In a first-author analysis of **11 LRDs at characteristic redshift $z \sim 4$** , I used medium- and high-resolution JWST/NIRSpec spectra to separate broad and narrow Balmer emission and examine its connection to the UV and optical continua (Pang et al. 2026a). This provides a spectroscopic route to disentangling the components behind their unusual spectral energy distributions.

The broad $H\alpha$ emission correlates strongly with the optical continuum but not with the UV, supporting a common AGN origin for the broad line and optical light. Conversely, $[O\text{ III}]$ emission tracks the UV continuum more closely. These different correlations indicate that the UV and optical emission need not arise from the same physical component; they do not, by themselves, uniquely establish the UV power source. The spectra also place upper limits on optical Fe II emission, providing an additional constraint on comparisons with ordinary AGN.

Subsequent studies have independently supported these line–continuum connections and the distinction between the red and blue components. In the NEXUS sample, Pan et al. (2026) confirm the broad- $H\alpha$ –optical-continuum correlation and find a much weaker connection between $[O\text{ III}]$ and the optical continuum. Complementary NIRSpec IFU observations by Ishikawa et al. (2026) spatially associate broad Balmer emission with compact red cores and narrow-line emission with blue host galaxies, strengthening the physical interpretation of our spectroscopic results.

Using broad- $H\alpha$ estimators, we infer black-hole masses of order 10^6 – $10^8 M_\odot$ and a characteristic Eddington ratio of $\lambda_{\text{Edd}} \sim 0.6$. The inferred growth timescales are 10^5 – 10^7 years, motivating a possible evolutionary connection to narrow-line Seyfert 1 galaxies.

An extended, clumpy gas shell. To reconcile the inferred broad-line-region scale with the optical luminosity, we proposed a **clumpy gas envelope extending over tens of light-days** (Pang et al. 2026a). Temperature and gas density decrease outward through the shell. Emission from layers at different temperatures and self-absorption by outer gas offer an explanation for the diversity of LRD optical continuum shapes. This work advances my research from identifying accreting black holes to testing their physical environments, and motivates combining line diagnostics with multi-epoch measurements to study how reprocessing shapes the observed emission.

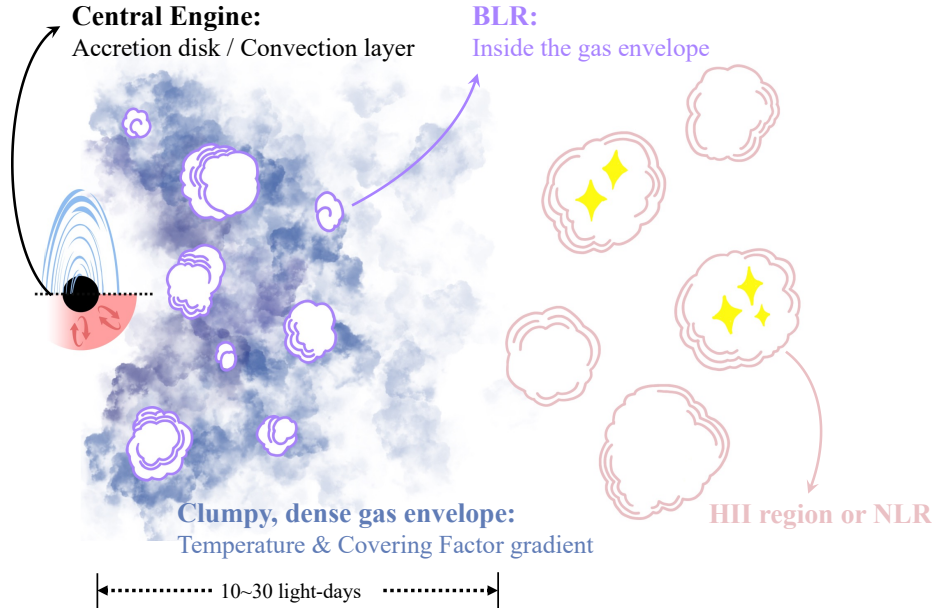


Figure 4. A physical interpretation of LRD spectra. The proposed clumpy-envelope model, reproduced from Figure 6 of Pang et al. (2026a). A shell extending from 10 to 30 light-days surrounds the central engine, with temperature, gas density, and layer covering factor decreasing outward. This is a schematic of the proposed structure.

4. TIME-DOMAIN EXTENSION: DISCOVERING AND CHARACTERIZING DISTANT TRANSIENTS

I have also extended my work to multi-epoch JWST imaging through the Pandora cluster Lensing, AGN, and Transient Exploration (PLATE) project. My first-author PLATE study reports **PLATE-23a, a Type II supernova candidate in a galaxy at $z = 0.688$** (Pang et al. 2026c). NIRCam observations from 2022–2025 detect the transient in 12 filters; difference imaging reveals both its rise and decline. Bayesian light-curve classification favors a Type IIP interpretation, and blackbody fits indicate cooling from approximately 8,010 to 6,100 K between the early epochs. Host-galaxy analysis places the event at a lensing-corrected projected separation of about 13 kpc from the center of a barred spiral. This study combines transient discovery, multiband modeling, and the resolved host environment; the classification remains photometric.

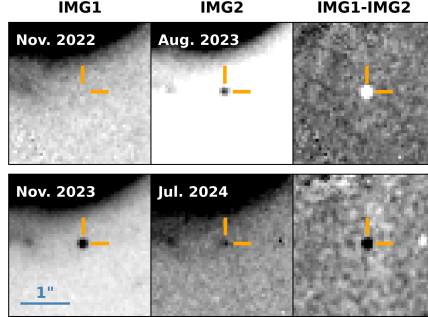


Figure 5. Opening the time dimension. F444W images and differences reveal the brightening (top) and fading (bottom) of PLATE-23a. Reproduced from Figure 1 of Pang et al. (2026c), currently an arXiv preprint.

In **PLATE II (in preparation)**, I am extending this work to PLATE-23b-1, a candidate microlensed red supergiant in a galaxy at $z \simeq 2.39$ behind Abell 2744. Combining multi-epoch NIRCam photometry, stellar SED modeling, and NIRSpec host-galaxy spectroscopy, this project uses gravitational magnification to probe individual cool, massive stars at cosmic noon.

5. RESEARCH EXPERIENCE AND DIRECTION

My first-author work brings together slitless-spectral forward modeling, machine learning, emission-line and SED fitting, survey-completeness analysis, and multi-epoch imaging. Collaborative projects have broadened this experience through quasar-catalog construction, optical spectroscopic follow-up, AGN variability, and interpretations of broad-line-region geometry. I have also taught spectral decomposition at the CSST Summer School and supervised undergraduate research, strengthening my ability to communicate methods and support collaborative analysis.

For my next fellowship, I aim to build on this combination of survey methods and physical diagnostics to study black-hole growth alongside galaxy evolution. A natural focus is to connect AGN samples selected through emission lines with JWST measurements of their continua, hosts, and variability. My HST and CSST work provides the selection framework; GLASS-JWST provides experience in population statistics; and the LRD and PLATE studies add physical modeling and time-domain diagnostics. Together, these form the basis for an independent research program that connects how sources are discovered with what can be inferred about their evolution.

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